

FundFlow — Fund Flow Dynamics

This diagram set explains how money moves through FundFlow: **two parallel pools** (cash and fund), kept in sync between master and member accounts, with **bank clearance** as a separate reconciliation step.

1. Account map — where money lives

flowchart TB

```
subgraph external["Outside the system"]
    BANK["Member's external bank account"]
end

subgraph master["Master pool (whole fund)"]
    MB["Master Bank<br/><i>imported statement lines</i>"]
    MC["Master Cash<br/><i>cash on hand</i>"]
    MF["Master Fund<br/><i>pool value</i>"]
end

subgraph member["Per member"]
    mC["Member Cash<br/><i>spendable balance</i>"]
    mF["Member Fund<br/><i>equity in pool</i>"]
    LA["Loan sub-account<br/><i>principal owed</i>"]
end

BANK <-->|"wire / transfer"| MB
MB <-->|"mirror to cash"| MC
MC <-->|"always mirrored"| mC
MF <-->|"always mirrored"| mF

style MC fill:#e8f4fc
style MF fill:#e8fce8
style mC fill:#e8f4fc
style mF fill:#e8fce8
```

Pool rules (checked nightly):

Invariant	Meaning
Master Cash = Σ Member Cash	Total cash in the pool equals the sum of all members' cash
Master Fund = Σ Member Fund	Total fund value equals the sum of all members' fund balances

Cash and fund are **separate ledgers**. A member can have cash without fund movement (and vice versa) until an operation links them (e.g. contribution collection moves cash $\rightarrow$ fund).

2. How money enters the system

Two main paths; both end with **member cash credited** (and master cash mirrored).

flowchart LR

```
subgraph pathA["Path A - Member deposit (Fund Posting)"]
```

```

    A1["Member submits deposit"] --> A2["Admin accepts"]
    A2 --> A3["CR Master Cash + CR Member Cash"]
    A3 --> A4["Uncleared bank line created"]
    A4 --> A5["Later: match to imported CSV"]
end

subgraph pathB["Path B - Bank statement first"]
    B1["Import CSV → Master Bank"] --> B2["Mirror to Master Cash"]
    B2 --> B3["Post to member → Member Cash"]
    B3 --> B4["Auto-collection may run"]
end

EXT["External bank"] --> pathA
EXT --> pathB

```

Important: Ledger posting records **intent** (money is in the member’s cash account). **Bank clearance** (matching an imported line or an uncleared placeholder) confirms it against the real bank — it does **not** post extra cash/fund legs when matched correctly.

3. Monthly contribution collection

When a member has cash (from a deposit or legacy import mirror), the collection cycle **moves cash into fund equity**.

flowchart TD

```

START["Member has cash balance"] --> TRIGGER["Cycle opens / deposit arrives /<br/>onMemberCashIncrease"]
TRIGGER --> CHECK{"Cash contribution due?"}
CHECK -->|Yes| FULL["Collect full amount"]
CHECK -->|Partial| PART["Collect available;<br/>remainder stays PENDING/OVERDUE"]
CHECK -->|No| WAIT["Wait for more cash"]

FULL --> JE["Journal (paired mirrors)"]
PART --> JE

JE --> DR1["DR Member Cash"]
DR1 --> DR2["DR Master Cash <i>(mirror)</i>"]
DR2 --> CR1["CR Member Fund"]
CR1 --> CR2["CR Master Fund <i>(mirror)</i>"]

CR2 --> DONE["Contribution COLLECTED<br/>Member fund balance ↑"]

```

Economic meaning: Cash leaves the member’s “wallet” and becomes **fund equity** (their share of the pool). Late fees, if any, also debit member + master cash.

4. Loan lifecycle — disbursement, cash-out, repayment

4a. Disbursement (ledger only — no bank payout yet)

flowchart TD

```

APP["Loan approved & disbursed"] --> SPLIT{"Funding split"}

SPLIT --> MP["Member portion<br/><i>from member's fund equity</i>"]
SPLIT --> MAST["Master portion<br/><i>from pool</i>"]

```

```

MP --> F1["DR Member Fund + DR Master Fund<br/><i>(mirror)</i>"]
MAST --> F2["DR Member Fund + DR Master Fund<br/><i>(mirror)</i>"]

F1 --> LOAN["Loan account: DR principal"]
F2 --> LOAN

LOAN --> CASH["CR Member Cash + CR Master Cash<br/><i>(cash payout mirror)</i>"]

CASH --> POOL["Cash now sits in member account<br/><i>not yet sent to their bank</i>"]

```

4b. Cash-out (actual bank transfer)

flowchart LR

```

REQ["Member requests cash-out"] --> ACC["Admin accepts"]
ACC --> DR["DR Member Cash + DR Master Cash"]
DR --> UNC["Uncleared bank transaction<br/><i>intent to pay out</i>"]
UNC --> MATCH["Later: Clear/Match against<br/>imported bank debit line"]

```

4c. Loan repayment (EMI / imported payment)

flowchart TD

```

PAY["Payment received<br/><i>deposit or legacy import</i>"] --> MIRROR["CR Member Cash + CR Master (")
MIRROR --> COLLECT["DR Member Cash + DR Master Cash<br/><i>(repayment collection)</i>"]
COLLECT --> FUND["CR Member Fund + CR Master Fund<br/><i>(mirror)</i>"]
FUND --> LOANCR["CR Loan sub-account<br/><i>principal reduced</i>"]
LOANCR --> CHECK{"Master portion + settlement<br/>threshold repaid?"}
CHECK -->|Yes| COMPLETE["Loan completed;<br/>guarantor may be released"]
CHECK -->|No| ACTIVE["Loan stays active"]

```

Repayment target: Master fund slice + settlement threshold (e.g. 16%). The member's own fund portion at disbursement is **equity**, not cash EMI — only the pool's share (+ settlement) is collected via repayments.

5. Special case — 100% member-fund loan (e.g. loan #143)

When the full loan is funded from the member's own fund (`member_portion` = full amount, `master_portion` = 0, no settlement due):

flowchart LR

```

D["Disbursement"] --> FUND["Member fund debited;<br/>principal applied on loan account"]
FUND --> CASH["Full amount credited to member cash"]
CASH --> OUT["Member cashes out"]
OUT --> DONE["No EMI schedule -<br/>loan marked completed<br/><i>nothing owed back to pool</i>"]

```

There is **no pool exposure**, so no ongoing installment obligation.

6. End-to-end lifecycle (one member, simplified)

sequenceDiagram

```

participant Ext as External bank
participant MB as Master Bank
participant MC as Master Cash
participant mC as Member Cash

```

participant mF as Member Fund
 participant MF as Master Fund
 participant Loan as Loan account

Note over Ext,MF: MONEY IN
 Ext->>MB: Transfer in
 MB->>MC: Mirror to cash
 MC->>MC: Post / accept deposit (mirrored)

Note over mC,MF: CONTRIBUTION
 mC->>mC: Debit (collection)
 MC->>MC: Debit (mirror)
 mF->>mF: Credit
 MF->>MF: Credit (mirror)

Note over mF,Loan: LOAN DISBURSE
 mF->>mF: Debit (member + master portions)
 MF->>MF: Debit (mirror)
 Loan->>Loan: Debit principal
 MC->>mC: Credit payout (mirror)

Note over mC,Ext: CASH OUT
 mC->>mC: Debit
 MC->>MC: Debit (mirror)
 mC-->>Ext: Bank transfer (cleared later)

Note over mC,Loan: REPAYMENT
 MC->>mC: Credit (mirror in)
 mC->>mC: Debit (EMI)
 MC->>MC: Debit (mirror)
 mF->>mF: Credit
 MF->>MF: Credit (mirror)
 Loan->>Loan: Credit principal

7. Mental model for stakeholders

Question	Answer
Where is “real” bank money?	Master Bank (imported statements) + uncleared lines until matched
What can a member spend?	Member Cash (after collection hasn’t taken it for contributions/EMIs)
What is their stake in the pool?	Member Fund (contributions – disbursements + repayments)
When does the pool shrink/grow?	Fund legs on contribution (in), disbursement (out), repayment (in)
Why two steps for bank?	Ledger = economic event; Clearance = proof it hit the bank

8. Operational functions → account touchpoints

Function	Cash legs	Fund legs	Bank
Deposit accept	CR member + master cash	—	Uncleared line → match
Bank import	Mirror → post to member	—	Master bank updated
Contribution	DR member + master cash	CR member + master fund	—
Loan disburse	CR member + master cash	DR member + master fund	—
Cash-out approve	DR member + master cash	—	Uncleared → match
Loan repayment	CR then DR cash (mirror pair)	CR member + master fund	—
Subscription fee	DR member + master cash	Fee to master fees	—
Legacy import	Same mirror patterns	Same mirror patterns	Often skipped